DNS

Installing dig.

```
ruupe@ruupe-debian:~$ sudo apt-get install dnsutils
[sudo] password for ruupe:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
The following additional packages will be installed:
  bind9-host bind9-libs libfstrm0 libjemalloc2 libmaxminddb0 libprotobuf-c1 libuv1t64
Suggested packages:
  mmdb-bin
The following NEW packages will be installed:
  bind9-dnsutils bind9-host bind9-libs libfstrm0 libjemalloc2 libmaxminddb0 libprotobuf-c1
  libuv1t64
0 upgraded, 8 newly installed, 0 to remove and 69 not upgraded.
Need to get 1 961 kB of archives.
After this operation, 5 832 kB of additional disk space will be used.
Do you want to continue? [Y/n]
Get:1 http://deb.debian.org/debian trixie/main amd64 libfstrm0 amd64 0.6.1-1+b3 [21.9 kB]
```

Testing DNS with dig

```
ruupe@ruupe-debian:~$ dig test034.linuxkurssi.xyz

; <<>> DiG 9.20.21-1~deb13u1-Debian <<>> test034.linuxkurssi.xyz
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 32843
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;test034.linuxkurssi.xyz.       IN      A

;; ANSWER SECTION:
test034.linuxkurssi.xyz. 1799   IN      A       65.52.68.180

;; Query time: 24 msec
;; SERVER: 192.168.100.1#53(192.168.100.1) (UDP)
;; WHEN: Wed Apr 01 23:38:37 EEST 2026
;; MSG SIZE  rcvd: 68

ruupe@ruupe-debian:~$
```

Running tcpdump on port 53 and running DNS query twice.

The difference on the 1st and 2nd query is that the ID's and source ports are randomized for each query while both returned the same IP address.

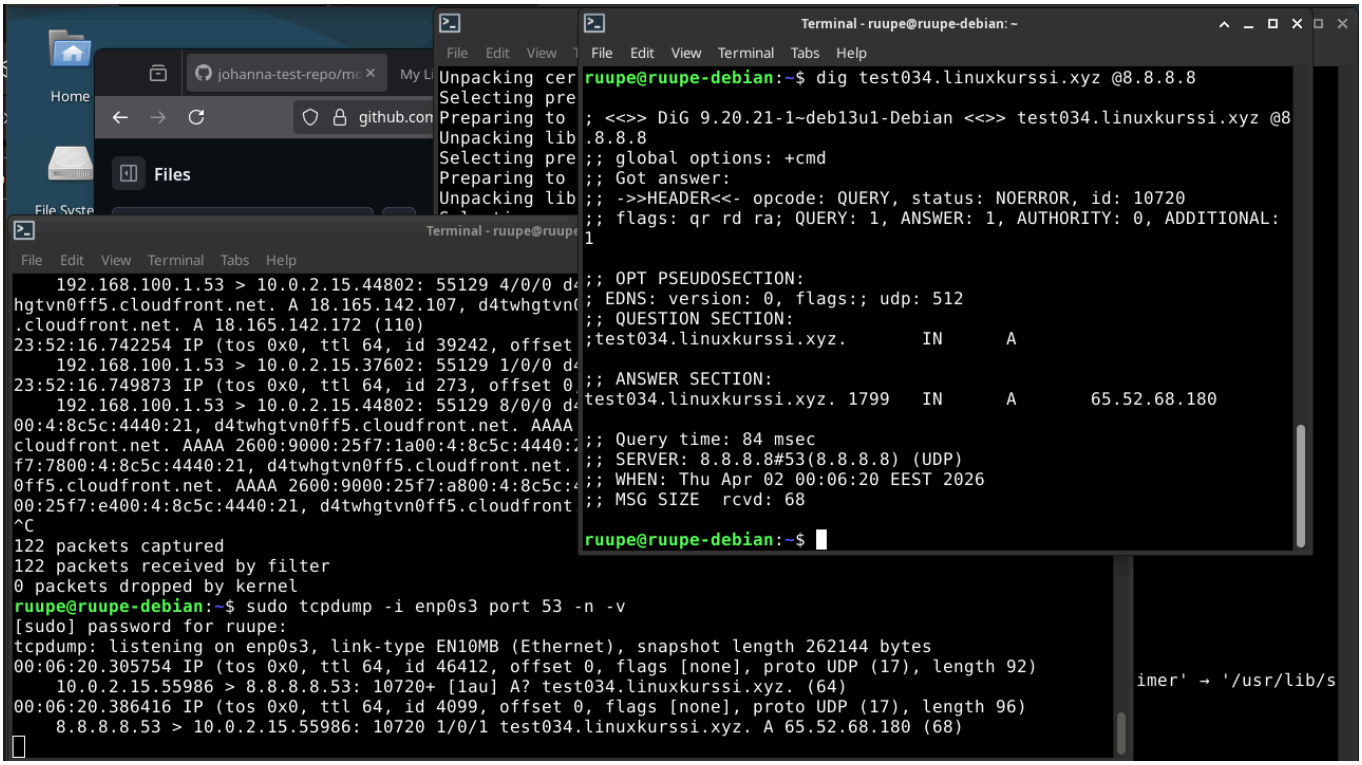
```
Terminal - ruupe@ruupe-debian: ~
File  Edit  View  Terminal  Tabs  Help

ruupe@ruupe-debian:~$ dig test034.linuxkurssi.xyz

; <<>> DiG 9.20.21-1~deb13u1-Debian <<>> test034.linuxkurssi.xyz
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 23285
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL:
1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;test034.linuxkurssi.xyz.       IN      A

;; ANSWER SECTION:
test034.linuxkurssi.xyz. 1082   IN      A       65.52.68.180

;; Query time: 4 msec
;; SERVER: 192.168.100.1#53(192.168.100.1) (UDP)
;; WHEN: Wed Apr 01 23:50:34 EEST 2026
;; MSG SIZE  rcvd: 68

ruupe@ruupe-debian:~$
```

```
ruupe@ruupe-debian:~$ sudo tcpdump -i enp0s3 port 53 -n -v
tcpdump: listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
23:50:34.171745 IP (tos 0x0, ttl 64, id 10673, offset 0, flags [none], proto UDP (17), length 92)
    10.0.2.15.40176 > 192.168.100.1.53: 23285+ [1au] A? test034.linuxkurssi.xyz. (64)
23:50:34.176472 IP (tos 0x0, ttl 64, id 39829, offset 0, flags [none], proto UDP (17), length 96)
    192.168.100.1.53 > 10.0.2.15.40176: 23285 1/0/1 test034.linuxkurssi.xyz. A 65.52.68.180 (68)
23:51:26.568000 IP (tos 0x0, ttl 64, id 43939, offset 0, flags [none], proto UDP (17), length 92)
    10.0.2.15.36127 > 192.168.100.1.53: 2672+ [1au] A? test034.linuxkurssi.xyz. (64)
23:51:26.578161 IP (tos 0x0, ttl 64, id 34997, offset 0, flags [none], proto UDP (17), length 96)
    192.168.100.1.53 > 10.0.2.15.36127: 2672 1/0/1 test034.linuxkurssi.xyz. A 65.52.68.180 (68)
```

Forcing the query to an external DNS server:

Using `@8.8.8.8` sends the query to an external DNS server, so the packet has to leave my VM and tcpdump can see it.

In earlier captures my queries went to 192.168.100.1 by default, but with `@8.8.8.8` they go to Google instead. So DNS is just UDP traffic on port 53, and I can choose which server to query.



Name-Based VirtualHost

Created the conf file for the namebased VirtualHost.

```
linuxuser@test034:/etc/apache2/sites-available$ more test034.linuxkurssi.xyz.conf
<VirtualHost *:80>
  ServerName test034.linuxkurssi.xyz
  ServerAlias www.test034.linuxkurssi.xyz
  DocumentRoot /home/linuxuser/public-sites/

  <Directory /home/linuxuser/public-sites/>
      Require all granted
  </Directory>

  Errorlog ${APACHE_LOG_DIR}/error-site1.log
  Customlog ${APACHE_LOG_DIR}/access-site1.log combined

</VirtualHost>
```

Created index.html file to /public-sites

```
linuxuser@test034:~/public-sites$ more index.html
<!DOCTYPE html>
<html>
<head><title>test034</title></head>
<body><h1>Hello from test034.linuxkurssi.xyz!</h1></body>
</html>
```

Created symbolic link from sites-available to sites-enabled

```
linuxuser@test034:/etc/apache2/sites-available$ sudo a2ensite test034.linuxkurssi.xyz
.conf
perl: warning: Setting locale failed.
perl: warning: Please check that your locale settings:
        LANGUAGE = (unset),
        LC_ALL = (unset),
        LC_CTYPE = (unset),
        LC_NUMERIC = "fi_FI.UTF-8",
        LC_COLLATE = (unset),
        LC_TIME = "fi_FI.UTF-8",
        LC_MESSAGES = (unset),
        LC_MONETARY = "fi_FI.UTF-8",
        LC_ADDRESS = "fi_FI.UTF-8",
        LC_IDENTIFICATION = "fi_FI.UTF-8",
        LC_MEASUREMENT = "fi_FI.UTF-8",
        LC_PAPER = "fi_FI.UTF-8",
        LC_TELEPHONE = "fi_FI.UTF-8",
        LC_NAME = "fi_FI.UTF-8",
        LANG = "C.UTF-8"
    are supported and installed on your system.
perl: warning: Falling back to a fallback locale ("C.UTF-8").
Enabling site test034.linuxkurssi.xyz.
To activate the new configuration, you need to run:
  systemctl reload apache2
linuxuser@test034:/etc/apache2/sites-available$
```

After that I granted Apache acces to the home directory.

```
linuxuser@test034:~$ chmod o+x /home/linuxuser/
linuxuser@test034:~$ curl http://test034.linuxkurssi.xyz
<!DOCTYPE html>
<html>
<head><title>test034</title></head>
<body><h1>Hello from test034.linuxkurssi.xyz!</h1></body>
</html>
linuxuser@test034:~$
```

Doing curl from outside the server gives 403.

```
ruupe@ruupe-debian:~$ curl -v http://test034.linuxkurssi.xyz
* Host test034.linuxkurssi.xyz:80 was resolved.
* IPv6: (none)
* IPv4: 65.52.68.180
*   Trying 65.52.68.180:80...
* Connected to test034.linuxkurssi.xyz (65.52.68.180) port 80
* using HTTP/1.x
> GET / HTTP/1.1
> Host: test034.linuxkurssi.xyz
> User-Agent: curl/8.14.1
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 403 Forbidden
< Server: F-Secure NIF
< Connection: close
< Content-Length: 0
<
* shutting down connection #0
```

I created a new sile with linuxuser. Then tried to switch to edituser. We did not have that user yet so I created it.

```
linuxuser@test034:~$ echo "<h1>file by linuxuser</h1>" > /home/linuxuser/public-sites/linuxuser.html
linuxuser@test034:~$ ls -l /home/linuxuser/public-sites/
total 8
-rw-rw-r-- 1 linuxuser linuxuser 125 Apr 22 07:53 index.html
-rw-rw-r-- 1 linuxuser linuxuser  27 Apr 22 08:41 linuxuser.html
linuxuser@test034:~$ su - edituser
su: user edituser does not exist or the user entry does not contain all the required fields
linuxuser@test034:~$ sudo adduser edituser
perl: warning: Setting locale failed.
perl: warning: Please check that your locale settings:
        LANGUAGE = (unset),
        LC_ALL = (unset),
        LC_CTYPE = (unset),
        LC_NUMERIC = "fi_FI.UTF-8",
        LC_COLLATE = (unset),
        LC_TIME = "fi_FI.UTF-8",
        LC_MESSAGES = (unset),
        LC_MONETARY = "fi_FI.UTF-8",
        LC_ADDRESS = "fi_FI.UTF-8",
        LC_IDENTIFICATION = "fi_FI.UTF-8",
        LC_MEASUREMENT = "fi_FI.UTF-8",
        LC_PAPER = "fi_FI.UTF-8",
        LC_TELEPHONE = "fi_FI.UTF-8",
        LC_NAME = "fi_FI.UTF-8",
        LANG = "C.UTF-8"
    are supported and installed on your system.
perl: warning: Falling back to a fallback locale ("C.UTF-8").
```

Now we can switch to edituser and create the file. But actually we can't beause we dont have the permission to do that.

```
edituser@test034:~$ echo "<h1>file by edituser</h1>" > /home/linuxuser/public-sites/edituser.html
-bash: /home/linuxuser/public-sites/edituser.html: Permission denied
edituser@test034:~$
```

We first had to give other group the write permissions

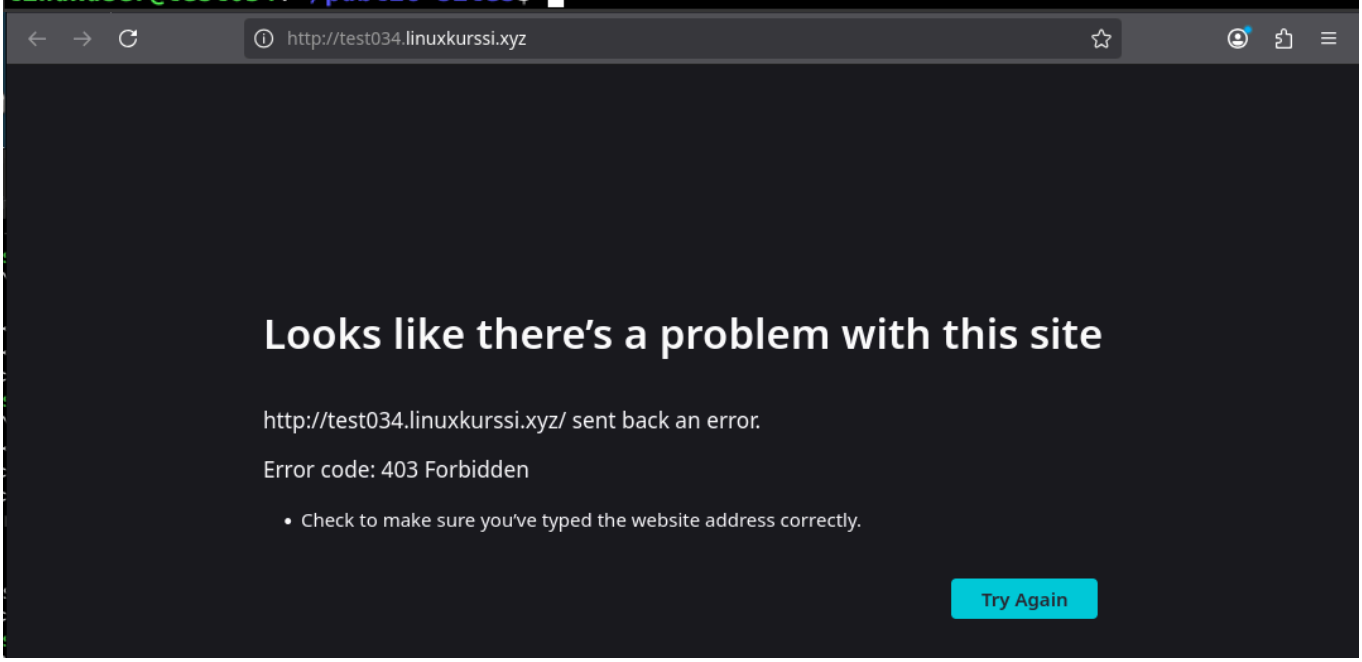
```
linuxuser@test034:~$ chmod o+w /home/linuxuser/public-sites/
linuxuser@test034:~$ ls -ld /home/linuxuser/public-sites/
drwxrwxrwx 2 linuxuser linuxuser 4096 Apr 22 08:41 /home/linuxuser/public-sites/
linuxuser@test034:~$
```

Only then we could create the file with "edituser"

```
edituser@test034:/home/linuxuser$ echo "<h1>file by edituser</h1>" > /home/linuxuser/public-sites/editu
ser.html
edituser@test034:/home/linuxuser$ ls -l /home/linuxuser/public-sites/
total 12
-rw-rw-r-- 1 edituser  edituser   26 Apr 22 10:15 edituser.html
-rw-rw-r-- 1 linuxuser linuxuser 125 Apr 22 07:53 index.html
-rw-rw-r-- 1 linuxuser linuxuser  27 Apr 22 08:41 linuxuser.html
```

Testing curl and browser. For some readon broswer still gives 403.

```
linuxuser@test034:~/public-sites$ curl http://test034.linuxkurssi.xyz
<!DOCTYPE html>
<html>
<head><title>test034</title></head>
<body><h1>Hello from test034.linuxkurssi.xyz!</h1></body>
</html>
linuxuser@test034:~/public-sites$
```



TLS Certificate

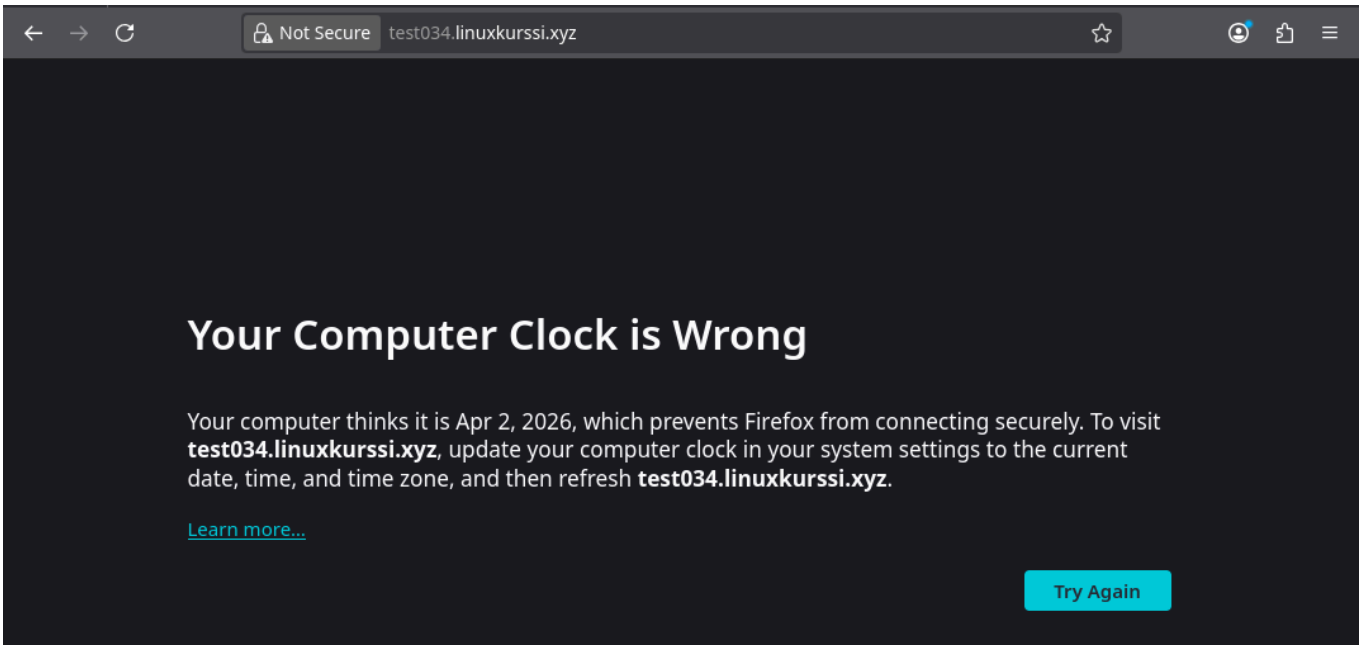
Installing certbot

```
linuxuser@test034:~$ sudo apt-get install certbot python3-certbot-apache
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  augeas-lenses libaugeas0 libicu76 python3-acme python3-augeas python3-certbot
  python3-configargparse python3-icu python3-josepy python3-openssl
  python3-parsedatetime python3-pytz python3-rfc3339
Suggested packages:
  augeas-doc python-certbot-doc python3-certbot-nginx augeas-tools python-acme-doc
  python-certbot-apache-doc python-openssl-doc
The following NEW packages will be installed:
  augeas-lenses certbot libaugeas0 libicu76 python3-acme python3-augeas
  python3-certbot python3-certbot-apache python3-configargparse python3-icu
  python3-josepy python3-openssl python3-parsedatetime python3-pytz python3-rfc3339
0 upgraded, 15 newly installed, 0 to remove and 0 not upgraded.
Need to get 11.9 MB of archives.
After this operation, 48.9 MB of additional disk space will be used.
Do you want to continue? [Y/n]
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [37 B]
Get:2 http://azure.deb.debian.cloud/debian trixie/main amd64 augeas-lenses all 1.14.1
-1.1~deb13u1 [463 kB]
Get:3 http://azure.deb.debian.cloud/debian trixie/main amd64 python3-openssl all 25.0
.0-1 [52.1 kB]
```

Creating TLS certificate

```
linuxuser@test034:~$ sudo certbot --apache -d test034.linuxkurssi.xyz,www.test034.linuxkurssi.xyz --reg
ister-unsafely-without-email
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for test034.linuxkurssi.xyz and www.test034.linuxkurssi.xyz

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/test034.linuxkurssi.xyz/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/test034.linuxkurssi.xyz/privkey.pem
This certificate expires on 2026-07-21.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for test034.linuxkurssi.xyz to /etc/apache2/sites-available/test034.l
inuxkurssi.xyz-le-ssl.conf
Successfully deployed certificate for www.test034.linuxkurssi.xyz to /etc/apache2/sites-available/test0
34.linuxkurssi.xyz-le-ssl.conf
Congratulations! You have successfully enabled HTTPS on https://test034.linuxkurssi.xyz and https://www
.test034.linuxkurssi.xyz

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt:   https://letsencrypt.org/donate
 * Donating to EFF:                    https://eff.org/donate-le
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
linuxuser@test034:~$
```

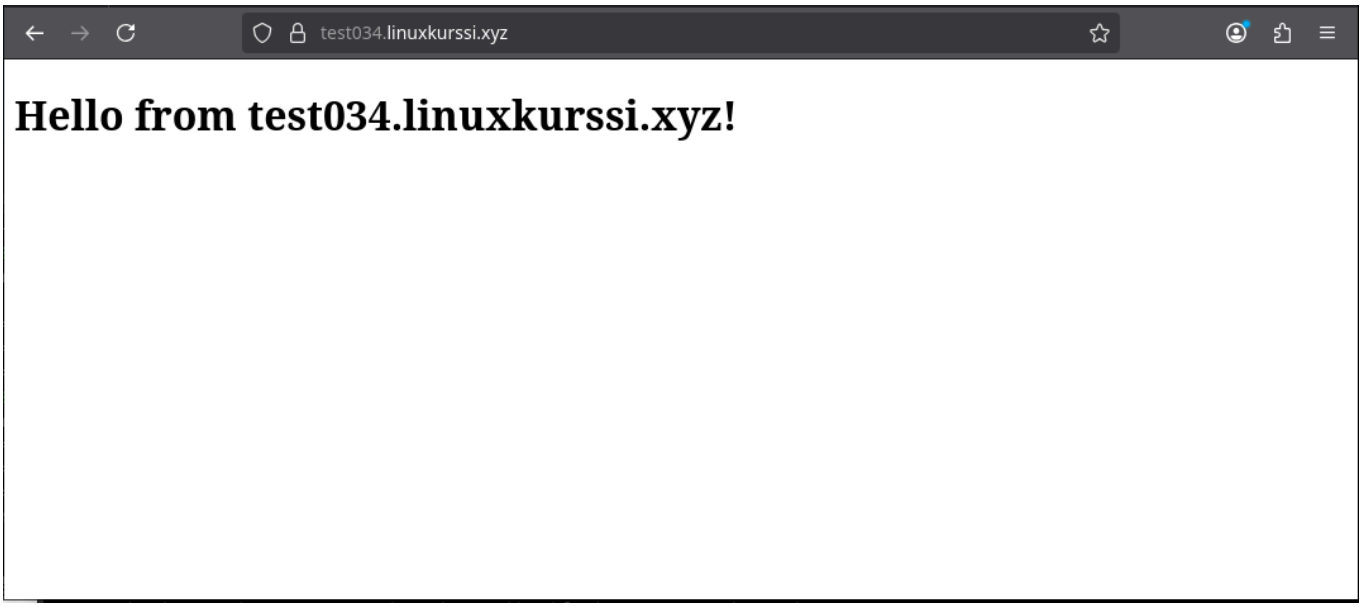
Testing with browser. This is something new. I need to figure out how to get my VM time corrected.



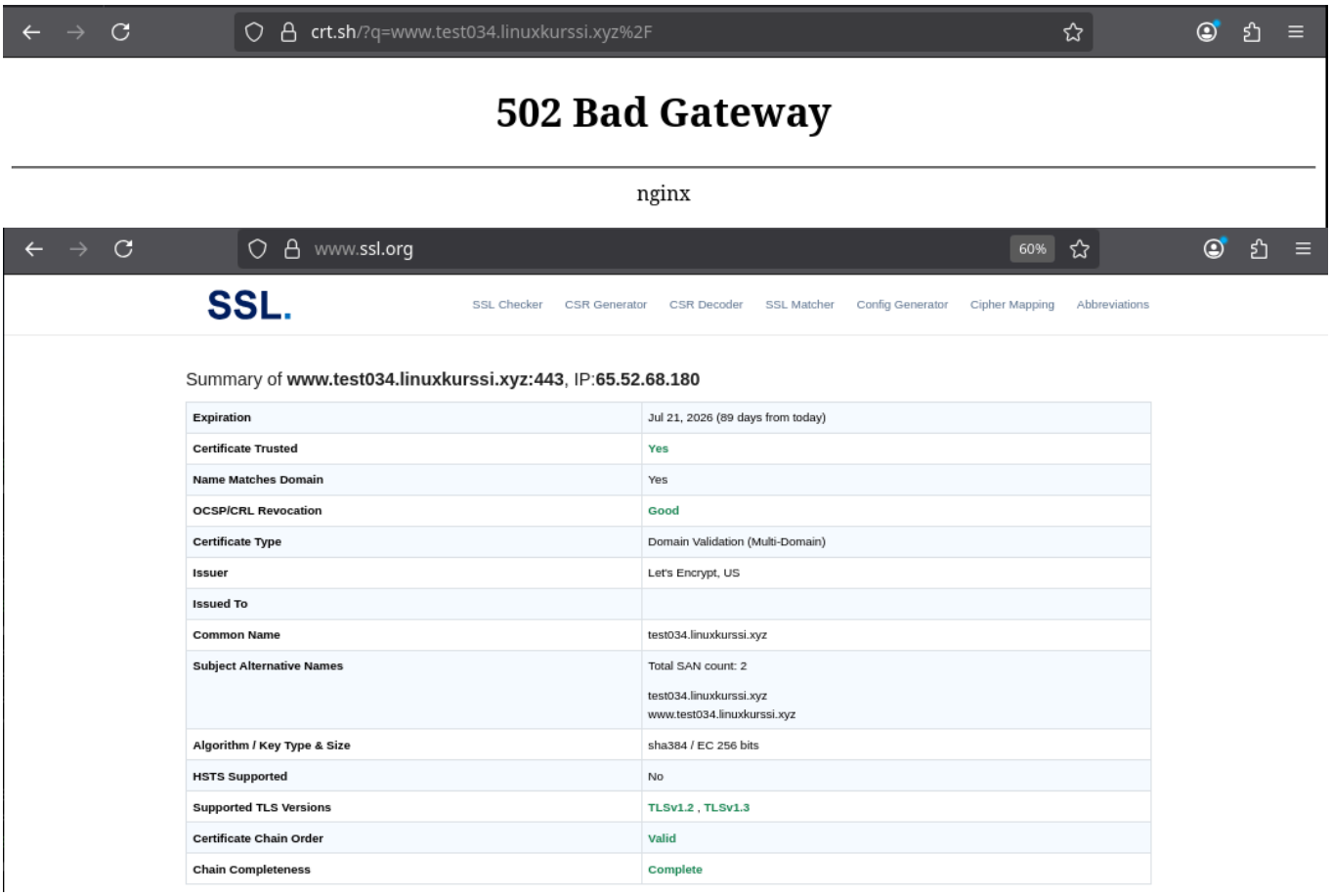
I fixed this by installing `systemd-timesyncd` and enabling automatic time synchronization with `sudo apt install systemd-timesyncd`

```
ruupe@ruupe-debian:~$ sudo apt install systemd-timesyncd
Installing:
  systemd-timesyncd

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 69
  Download size: 93,0 kB
  Space needed: 207 kB / 11,4 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 systemd-timesyncd amd64 257.9-1~deb13u1 [93,0 kB]
Fetched 93,0 kB in 0s (1 011 kB/s)
Selecting previously unselected package systemd-timesyncd.
(Reading database ... 218925 files and directories currently instal
led.)
Preparing to unpack .../systemd-timesyncd_257.9-1~deb13u1_amd64.deb
 ...
Unpacking systemd-timesyncd (257.9-1~deb13u1) ...
Setting up systemd-timesyncd (257.9-1~deb13u1) ...
Creating group 'systemd-timesync' with GID 989.
Creating user 'systemd-timesync' (systemd Time Synchronization) wit
h UID 989 and GID 989.
Created symlink '/etc/systemd/system/dbus-org.freedesktop.timesync1
.service' → '/usr/lib/systemd/system/systemd-timesyncd.service'.
Created symlink '/etc/systemd/system/sysinit.target.wants/systemd-t
imesyncd.service' → '/usr/lib/systemd/system/systemd-timesyncd.serv
ice'.
systemd-time-wait-sync.service is a disabled or a static unit, not
starting it.
Processing triggers for dbus (1.16.2-2) ...
Processing triggers for man-db (2.13.1-1) ...
ruupe@ruupe-debian:~$ sudo timedatectl set-ntp true
[sudo] password for ruupe:
ruupe@ruupe-debian:~$
```

Testing with browser again. Success!



crt.sh seems to have some problems while other similar websites work. TLS is active on the website.



Simulating certificate renewal

```
linuxuser@test034:~/public-sites$ sudo certbot renew --dry-run
Saving debug log to /var/log/letsencrypt/letsencrypt.log

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Processing /etc/letsencrypt/renewal/test034.linuxkurssi.xyz.conf
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Account registered.
Simulating renewal of an existing certificate for test034.linuxkurssi.xyz and www.test034.linuxkurssi.xyz

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Congratulations, all simulated renewals succeeded:
  /etc/letsencrypt/live/test034.linuxkurssi.xyz/fullchain.pem (success)
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
linuxuser@test034:~/public-sites$
```

Monitoring – curl in verbose mode

Monitoring HTTPS traffic:

With HTTPS, curl shows the TLS handshake, certificate detail, and the decrypted page content.

```
ruupe@ruupe-debian:~$ curl -v https://test034.linuxkurssi.xyz
* Host test034.linuxkurssi.xyz:443 was resolved.
* IPv6: (none)
* IPv4: 65.52.68.180
*   Trying 65.52.68.180:443...
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
*  CAfile: /etc/ssl/certs/ca-certificates.crt
*  CApath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (IN), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
* TLSv1.3 (IN), TLS handshake, Certificate (11):
* TLSv1.3 (IN), TLS handshake, CERT verify (15):
* TLSv1.3 (IN), TLS handshake, Finished (20):
* TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (OUT), TLS handshake, Finished (20):
* SSL connection using TLSv1.3 / TLS_AES_256_GCM_SHA384 / X25519MLKEM768 / id-ecPublicKey
* ALPN: server accepted http/1.1
* Server certificate:
*  subject: CN=test034.linuxkurssi.xyz
*  start date: Apr 22 09:23:40 2026 GMT
*  expire date: Jul 21 09:23:39 2026 GMT
*  subjectAltName: host "test034.linuxkurssi.xyz" matched cert's "test034.linuxkurssi.xyz"
*  issuer: C=US; O=Let's Encrypt; CN=E7
*  SSL certificate verify ok.
*   Certificate level 0: Public key type EC/prime256v1 (256/128 Bits/secBits), signed using ecdsa-with-SHA384
*   Certificate level 1: Public key type EC/secp384r1 (384/192 Bits/secBits), signed using sha256WithRSAEncryption
*   Certificate level 2: Public key type RSA (4096/152 Bits/secBits), signed using sha256WithRSAEncryption
* Connected to test034.linuxkurssi.xyz (65.52.68.180) port 443
* using HTTP/1.x
> GET / HTTP/1.1
> Host: test034.linuxkurssi.xyz
> User-Agent: curl/8.14.1
> Accept: */*
>
* Request completely sent off
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
< HTTP/1.1 200 OK
< Date: Wed, 22 Apr 2026 10:56:19 GMT
< Server: Apache/2.4.66 (Debian)
< Last-Modified: Wed, 22 Apr 2026 07:53:23 GMT
< ETag: "7d-65007d37a13b4"
< Accept-Ranges: bytes
< Content-Length: 125
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html>
<head><title>test034</title></head>
<body><h1>Hello from test034.linuxkurssi.xyz!</h1></body>
</html>
* Connection #0 to host test034.linuxkurssi.xyz left intact
ruupe@ruupe-debian:~$
```

Monitoring HTTP traffic:

When monitoring HTTP traffic, the request was blocked by F-Secure NIF with a 403 Forbidden response before even reaching the Apache server.

```
ruupe@ruupe-debian:~$ curl -v http://test034.linuxkurssi.xyz
* Host test034.linuxkurssi.xyz:80 was resolved.
* IPv6: (none)
* IPv4: 65.52.68.180
*   Trying 65.52.68.180:80...
* Connected to test034.linuxkurssi.xyz (65.52.68.180) port 80
* using HTTP/1.x
> GET / HTTP/1.1
> Host: test034.linuxkurssi.xyz
> User-Agent: curl/8.14.1
> Accept: */*
>
< HTTP/1.1 403 Forbidden
< Server: F-Secure NIF
< Connection: close
< Content-Length: 0
<
* we are done reading and this is set to close, stop send
* abort upload
* shutting down connection #0
ruupe@ruupe-debian:~$
```

After commenting out the "Rewrite" lines, restarting Apache and using curl on the HTTP address I got this reponse.

```
linuxuser@test034:/etc/apache2/sites-available$ curl -v http://test034.linuxkurssi.xyz
* Host test034.linuxkurssi.xyz:80 was resolved.
* IPv6: (none)
* IPv4: 65.52.68.180
*   Trying 65.52.68.180:80...
* Connected to test034.linuxkurssi.xyz (65.52.68.180) port 80
* using HTTP/1.x
> GET / HTTP/1.1
> Host: test034.linuxkurssi.xyz
> User-Agent: curl/8.14.1
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Wed, 22 Apr 2026 11:25:01 GMT
< Server: Apache/2.4.66 (Debian)
< Last-Modified: Wed, 22 Apr 2026 07:53:23 GMT
< ETag: "7d-65007d37a13b4"
< Accept-Ranges: bytes
< Content-Length: 125
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html>
<head><title>test034</title></head>
<body><h1>Hello from test034.linuxkurssi.xyz!</h1></body>
</html>
* Connection #0 to host test034.linuxkurssi.xyz left intact
linuxuser@test034:/etc/apache2/sites-available$
/home/ruupe
```

After putting rewrites on again:

```
linuxuser@test034:/etc/apache2/sites-available$ curl -v http://test034.linuxkurssi.xyz
* Host test034.linuxkurssi.xyz:80 was resolved.
* IPv6: (none)
* IPv4: 65.52.68.180
*   Trying 65.52.68.180:80...
* Connected to test034.linuxkurssi.xyz (65.52.68.180) port 80
* using HTTP/1.x
> GET / HTTP/1.1
> Host: test034.linuxkurssi.xyz
> User-Agent: curl/8.14.1
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 301 Moved Permanently
< Date: Wed, 22 Apr 2026 11:26:49 GMT
< Server: Apache/2.4.66 (Debian)
< Location: https://test034.linuxkurssi.xyz/
< Content-Length: 369
< Content-Type: text/html; charset=iso-8859-1
<
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
<p>The document has moved <a href="https://test034.linuxkurssi.xyz/">here</a>.</p>
<hr>
<address>Apache/2.4.66 (Debian) Server at test034.linuxkurssi.xyz Port 80</address>
</body></html>
* Connection #0 to host test034.linuxkurssi.xyz left intact
linuxuser@test034:/etc/apache2/sites-available$
```

With the redirect enabled, HTTP requests get a 301 Moved Permanently response pointing to the HTTPS version.

With the redirect disabled, HTTP returns 200 OK and serves the page in plain text.

In short, the Rewrite rules force the HTTP request to use the secure HTTPS protocol.